

Indian Institute of Technology Hyderabad
Maths for Physics (EP1118)
 Mid Term test, Date: 1st September, 2025

Timing: 12:00 AM to 1:30 PM

Max mark: 40

Section A: Each question carries 10 marks. (Any 4). NO STEP JUMP.
 First 4 questions will be checked

- ✓ 1. (i) If R is a closed region of the xy plane bounded by a simple closed curve C , and if M and N are continuous functions of x and y having continuous derivatives in R , then prove Green's theorem, i.e.,

$$\oint_C M(x, y)dx + N(x, y)dy = \iint_R \left(\frac{\partial N(x, y)}{\partial x} - \frac{\partial M(x, y)}{\partial y} \right) dx dy \quad 3 \quad +10$$

- (ii) Using Green's theorem, find the area of the region in the first quadrant bounded by the curves $y = x$, $y = \frac{1}{x}$, $y = \frac{x}{4}$. 4

- (iii) A vector field $\vec{F}$ is given by $\vec{F} = \sin y \hat{i} + x(1 + \cos y) \hat{j}$. Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$, where C is the circular path given by $x^2 + y^2 = a^2$. 3

- ✓ 2. (i) **Prove the divergence theorem** where the surface integral of the normal component of a vector function $\vec{F}$ taken around a closed surface S is equal to the integral of the divergence of $\vec{F}$ taken over the volume V enclosed by the surface S . Mathematically

$$\iiint_V \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv \quad 6 \quad +7$$

- (ii) Evaluate surface integral $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = (x^2 + y^2 + z^2)(\hat{i} + \hat{j} + \hat{k})$, S is the surface of the tetrahedron $x = 0, y = 0, z = 0, x + y + z = 2$ and $\hat{n}$ is the unit normal in the outward direction to the closed surface S . 4

3. (i) Describe the cylindrical coordinate with a proper picture. And how we can write dx, dy, dz in cylindrical co-ordinates. 2+2

- (ii) If the position vector satisfies $\vec{r} \equiv r(x, y, z) \equiv r(\rho, \phi, Z)$, then what would be the $\hat{\rho}, \hat{\phi}$ and $\hat{z}$ in terms of $\hat{i}, \hat{j}, \hat{k}$. 3

- (iii) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = y^2 \hat{i} + x^2 \hat{j} - (x + z) \hat{k}$ and C is the boundary of the triangle with vertices at $(0, 0, 0), (1, 0, 0)$, and $(1, 1, 0)$. 3

4. (i) Write $\nabla, \nabla \times$ and ∇^2 in cylindrical co-ordinates. 4

- (ii) If the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then what would be the velocity and acceleration in the cylindrical coordinate system? 3+3

- ✓ 5. (i) Verify the Divergence theorem for the function $\vec{F} = 2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$ taken over the region in the first octant bounded by $y^2 + z^2 = 9$ and $x=2$. 5

- (ii) Verify Stoke's theorem for $\vec{F} = (x^2 + y - 4)\hat{i} + 3xy\hat{j} + (xz + z^2)\hat{k}$ over the surface of hemisphere $x^2 + y^2 + z^2 = 16$ above the $x - y$ plane. 5

- ✓ 6. (i) Prove $\oint \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} ds$. 7

- (ii) Apply the Divergence theorem to compute $\iint_S \vec{u} \cdot \hat{n} ds$, where S is the surface of the cylinder $x^2 + y^2 = a^2$ bounded by the planes $z = 0, z = b$ and where $\vec{u} = x\hat{i} - y\hat{j} + z\hat{k}$. 3